

Economy Exploit Report

HIGH ROLEPLAY server — crafting & selling systems review

Generated 16 July 2026 · Sources: ECONOMY GENERATOR HIGH RP (config) + high-roleplay-v1 (server code)

The short version

Your old economy was safe because **crafting took time**. Time was the payment. A player who refined iron for one hour earned about one hour of wage. Today crafting is **instant**, so every recipe that ends with a profit is now a **money printer**: click, click, click — unlimited money. But the review found something even more dangerous: **the server lets the player's game client choose the selling price**. A player with a modified client can sell one cheap item for any price he wants — for example 999,999,999 — and the server will pay it. This is worse than any price problem, and it must be fixed first.

Contents

- 1. How money is supposed to work on your server
- 2. The three holes in the server code (most important)
- 3. The money printers — every exploit, with real numbers
- 4. Broken price ladders (prices that contradict each other)
- 5. Systems that are OK (or low risk)
- 6. A hidden trap: your three configs do not match
- 7. The fix plan — step by step
- 8. Golden rules to keep the economy safe forever
- 9. Checklist

1. How money is supposed to work on your server

To judge an exploit, we need a ruler. Your ruler is the **hourly wage**:

Way of earning	Money per hour
Government job (police, EMS...) — base salary	~14,000
Electric job (level 1 → 3)	~9,000 – 17,000
Honest mining + refining (time-gated)	~20,000 – 50,000

So a "normal" hour of play is worth roughly **10,000 – 50,000**. Every exploit below should be compared with this number. When a loop makes 500,000+ per hour, salaries, jobs and normal prices stop meaning anything.

2. The three holes in the server code

These are in `high-roleplay-v1`, not in the price config. Prices cannot fix them — the code must change.

Hole #1 — Crafting time is completely ignored

- Every recipe in the config has `craftTimeMinutes` (3 min, 5 min, even 180 min for the level-4 pickaxe).
- **No line of server code ever reads this field.** The craft handler (`craft/craftItem.ts`) removes the materials and gives the finished item **in the same moment**.
- The old timer system still exists in the code — the database model (`crafting_model.ts`), the "collect finished item" handler (`collectItems.ts`), and the "reduce time" handler — but nothing connects to them anymore. It is dead code.
- Extra detail: in the deployed config the gang weapon recipes even have `craftTimeMinutes: 0` written by hand.

Hole #2 — The player's client chooses the selling price (critical)

- All NPC selling (drug dealer, weapon dealer, mining shop, EMS shop) goes through one handler: `server:sell:items` in `SellService.ts`.
- The handler receives `price` **from the client** and pays it with `moneyService.addMoney(player, price)`. It only checks that the number is not negative.
- The only "protection" is a 1-second cooldown **on the client side**, which a cheater simply deletes.
- **Meaning:** a player with a modified client does not need any crafting exploit. He can sell one 150\$ water bottle for 50,000,000\$ and the server pays. If your server has ever had unexplained rich players, this is the most likely door they used.

Hole #3 — The client also chooses the crafting recipe

- For mining, refinery, EMS and drug crafting, the client sends `costItems` (what to spend) and `itemName` (what to receive). The server checks that the player owns the materials — but **never checks that this is a real recipe**.
- A modified client can say: "take 1 stone (22\$) and give me an MDMA pill" — and the server obeys.

- Only the gang **weapon** bench is safe: it reads the real recipe from the server config, checks permissions, and caps the amount at 20.

Why this matters more than prices

Even if every price in this report is corrected, holes #2 and #3 still allow unlimited money with a modified client. Fix the code first, then the prices.

3. The money printers — every exploit, with real numbers

These loops need **no cheating at all** — a normal player with the normal game can do them today, because crafting is instant and NPCs both sell the materials and buy the product.

#1 Iron bars — buy ore, refine, sell back (your own example)

Step	Details	Money
Buy from mining shop	100 iron ore × 37 + 10 coal × 35	−4,050
Refine (should take 3 min, is instant)	→ 4 iron bars	
Sell to the same shop	4 bars × 1,512	+6,048
Profit per click	+49%	+1,998

One craft every 5 seconds ≈ **1,400,000 per hour** — one hundred times a police salary. The same shop sells the input **and** buys the output, so the loop never leaves the NPC. Gold bars are a second printer: cost 7,130 → sell 8,640 (**+1,510, +21%** per click).

#2 MDMA pill — the biggest drug printer

Item (drug dealer pays)	Cost to make	Dealer pays	Profit
Methylamine (3 methanol + 2 ammonia — all bought from supermarket)	4,800	6,554	+1,754 (+37%)
MDP2P (acetone + saffarole oil + ammonia)	~2,200	5,979	+3,777 (+172%)
MDMA pill (1 powder + 1 capsule 250\$)	~4,900	10,925	+6,007 (+122%)

- **Methylamine is the purest printer:** every material comes from the supermarket NPC. No farming, no risk, no time. ~630,000/hour.
- The old design notes in your own config say an MDMA pill should be worth about **4,250**. The dealer pays **10,925** — 2.5× too much.
- Cocaine powder: materials ~5,968 (including the value of the coca leaves) → dealer pays 14,490 for the batch = **+8,522 (+143%)** per craft. It is gated by gathering coca leaves, but with instant crafting the whole chain moves at gathering speed with a huge markup.

#3 Crafted weapons sold to the weapon dealer

- Generator recipe: knife = 5 iron bars. Bars refined from **bought** ore cost ~1,012 each → knife costs ~5,063 → dealer pays **7,000** = **+1,938 per knife, instant**.
- On the live server weapons are built from `item_military_component` (weapon-shop robbery loot). The dealer prices turn components into instant cash: the **knife is the best exchange rate (467\$ per component)** — better than any gun. So gangs will spam-craft knives to liquidate robbery loot. Craft time on these recipes is hand-set to **0**.
- The dealer also still lists prices for weapon parts (barrels, triggers...) that no longer exist as items — dead entries that become an exploit if those items ever return.

#4 EMS medical crafting — profitable, "protected" by limits that do not work

Chain (all materials from NPC shops)	Cost	EMS shop pays	Profit per craft
Bandage ×5 (3 ethanol + 5 fabric)	3,400	4,500	+1,100 (+32%)
Painkiller pill (per pill)	1,250	1,550	+300 (+24%)
First-aid kit (self-crafted materials)	~4,910	6,500	+1,590

Your config already contains sell limits for exactly this (`limitation: 8 bandages / 15 min`, fabric buy limit 15 / 30 min, price-decay fields). **The server never reads any of them.** With no limits and instant crafting, this is another (smaller) printer: roughly 30,000–60,000/hour with zero risk, better than any salary.

#5 Ammo crafting (gang)

1 iron bar + 8 sulfur (≈2,340 if ore is bought) → **200 bullets**, instantly. The weapon shop sells 9mm at 30\$; crafted cost is ~12\$. No NPC buys bullets back (good), so this is not a direct printer — but it floods the player market with almost-free ammo and kills the weapon shop's ammo business. With a 1-minute craft time this was fine; instant, it is a problem.

4. Broken price ladders

Even ignoring exploits, some prices contradict each other. Players always find these.

Problem	Numbers	Rule broken
Cocaine pack destroys value	3 powder (14,490) + 3 plastic (1,299) → 3 packs sell for only 15,000	A finished product must be worth more than its parts. Nobody will ever craft packs; the recipe is dead content.
MDMA powder is worth less than its ingredients	Inputs sell for ~8,355 (per powder made); powder sells 4,485	Same rule. Players sell the intermediates and skip the "final" product.
MDMA pill jumps 2.4×	Powder 4,485 + capsule 250 → pill 10,925	One cheap capsule should not multiply value by 2.4.
Old design price vs dealer price	Config comments say pill ≈ 4,250; dealer pays 10,925	The dealer file was tuned separately from the recipes file.
Weapon dealer buys ghosts	Prices exist for barrels/triggers/grips that can no longer be crafted or found	Remove dead entries before they become an exploit.

5. Systems that are OK (or low risk)

System	Verdict	Why
Mining itself (pickaxe hits)	OK	Time-gated: 4 s per hit, kits and repairs cost money. This is honest wage design.
Gilded iron / Auriron / gems	OK	Needs gems (0.06% drop). Selling raw gems pays slightly more than refining them — small loss, no printer.
Buy bar → sell bar directly	OK	Shop sells at 2,100, buys at 1,512. Correct direction.
Burger job	Low risk — verify	Ingredients are cheap (~250 per burger) and no NPC in the config buys finished burgers. But its bench recipes have their own <code>craftTime</code> field — make sure the burger benches are not also instant, and confirm how NPC orders pay.
Junk collecting	Low risk — verify	Prices are small and items come from searching bins. Risk depends only on bin respawn speed.

Booze / vending / gas station / supermarket food	OK	Buy-only, no sell-back loop.
First-aid kit price logic	Good design	The comment in <code>EMS.js</code> shows the right thinking: shop-bought materials cost 7,300 > sell price 6,500, so buy→craft→sell-back loses money. This rule should be applied everywhere.

6. A hidden trap: your three configs do not match

There are three copies of the economy, and all three are different right now:

- **Generator source** (`ECONOMY GENERATOR HIGH RP/*.js`) — has old gang weapon recipes (iron bars, barrels...) that no longer exist in the game, and has the first-aid-kit recipe and the reinforced-steel refinery move.
- **Generator output** (`economyConfig.json`, built 14 July) — stale: missing the reinforced-steel refinery entry.
- **Deployed config** (`high-roleplay-v1/.../shared/economyConfig.json`, edited 15 July) — gang weapons were rewritten **by hand** to use `item_military_component` with craft time 0, and it is missing the first-aid recipe.

Warning before you regenerate

If you run `npm run generate` today and copy the result to the server, you will **delete the military-component weapon recipes** and bring back recipes for items that do not exist. First port the military recipes into `CRAFTS/GANG.js`, then regenerate, then copy to **both** consumers (high-ui and high-roleplay-v1).

7. The fix plan — step by step

Step 1 — Close the code holes (do this first, prices cannot wait for anything else)

1. **Server-side prices.** In `SellService.ts`, ignore the price sent by the client. Look the price up in the server's own economy config by `itemName`. If the item is not in a sell list, refuse the sale. This one change kills the "sell water for a billion" hack.
2. **Server-side recipes.** In `craftItem.ts`, do what the gang weapon bench already does — for **every** craft type: find the recipe in the server config, take the materials the **recipe** says (not what the client says), give the output the recipe says. Cap the amount per request.
3. **Bring back crafting time.** The pieces already exist: the `Crafting` database model with `craftDuration`, the collect handler, the status handler. Reconnect them: when a craft starts, write a document with `finishesAt = now + craftTimeMinutes`; the player collects when the time has passed. Because it is stored in the database, relogging or server restart does not skip the timer. Set the gang-weapon recipes' hand-edited `craftTimeMinutes: 0` back to real values (3–5 min).
4. **Enforce the sell limits that already exist in the config** (count per time window, price decay after the limit). Apply them per player, per item, on the server, at every NPC that buys: drug dealer, weapon dealer, mining shop, EMS shop.
5. **Log every sale and craft** (player, item, amount, price, timestamp) and build a simple daily report: top earners, top items sold. An admin alert when one player sells more than e.g. 200,000 in an hour will catch every future exploit early.

Step 2 — Fix the prices (after Step 1, in the generator, then regenerate)

The one rule that makes NPC loops safe: **if all materials can be bought from NPCs, the NPC sell price of the product must be LOWER than the material cost**. Profit should only exist where the player spent time or risk (mining the ore himself, growing the leaves, robbing the weapon shop). Concrete suggestions:

Price	Now	Suggested	Effect
Iron ore — shop buy price	37	60	100 bought ore + coal = 6,350 > 6,048 bar revenue → bought-ore refining loses money; mined ore still earns ~6,000 per 100 ore of mining time.
Gold ore — shop buy price	53	75	Same logic: bought-ore gold bars become a small loss; mined gold stays profitable.
Methylamine — dealer pays	6,554	4,000	All-NPC chain becomes a loss (cost 4,800). It stays valuable as an ingredient for MDMA.

MDP2P — dealer pays	5,979	3,500	Still ~+1,300 profit per unit — but that profit now pays for growing saffarole leaves, not for clicking.
Cocaine powder — dealer pays	4,830	3,800	Profit $\approx$ leaf-gathering wage, not a printer.
Cocaine pack — dealer pays	5,000	4,600	With powder at 3,800, the pack is finally worth more than its parts.
MDMA powder — dealer pays	4,485	5,500	Worth more than its ingredients (5,000) — the ladder points upward again.
MDMA pill — dealer pays	10,925	6,200	$\sim +8\%$ over powder + capsule, instead of $+122\%$.
Weapon dealer — dead part entries	listed	remove	Barrels/triggers/grips/scopes/stocks no longer exist as items.
Knife — dealer pays (military-comp world)	7,000 (467/comp)	$\leq 5,250$ (350/comp)	Knife stops being the best way to liquidate robbery loot; guns stay on top.

Keep the EMS numbers as they are — they were designed correctly — but they only work once the limits are actually enforced (Step 1.4).

Step 3 — Protect the future

- **One source of truth.** Never hand-edit the deployed `economyConfig.json` again. Port the military recipes into the generator now, and make every future change generator-first.
- **Automatic exploit check.** Add a small script to the generator (run it inside `npm run generate`) that walks every recipe, computes material cost from NPC buy prices, compares with every NPC sell price, and **fails the build** if any all-NPC chain is profitable. This makes the #1 mistake impossible to repeat.
- **Craft time as design, not decoration.** Decide the wage you want per activity (e.g. refining $\approx 30,000/\text{hour}$) and set `craftTimeMinutes` so $\text{profit} \div \text{time} \approx$ that wage. Example: iron bar refine profit for mined ore $\approx 2,000$ per craft $\rightarrow 4$ minutes per craft $\approx 30,000/\text{hour}$.

8. Golden rules to keep the economy safe forever

1. **Never trust the client.** Prices, recipes, amounts — always from the server config.
2. **No profitable NPC circles.** If NPCs sell all the materials, NPCs must pay less than the material cost for the product.
3. **Profit = time or risk.** Money should only appear where the player spent minutes (mining, growing, driving) or took risk (robbery, drug runs near police).

4. **Ladders go up.** Every crafted product must be worth more than the sum of its parts — and intermediate items must be worth less than what they build.
5. **Everything sellable has a cap.** Per-player, per-item, per-time-window limits with price decay, enforced on the server.
6. **Watch the money.** Log every transaction; review the top 10 earners daily. Exploits show up in data long before players report them.
7. **One source of truth.** Generator → generate → copy to both consumers. Never edit generated JSON by hand.

9. Checklist

#	Task	Where	Priority
1	Sell price from server config, not client	SellService.ts	CRITICAL
2	Validate every craft against server recipes	craftItem.ts	CRITICAL
3	Reconnect craft timers (DB-backed, collect when done)	craftItem.ts + crafting_model.ts	HIGH
4	Enforce sell/buy limits + price decay at all NPC buyers	SellService.ts	HIGH
5	Transaction logging + daily top-earner report	server	HIGH
6	Port military-component recipes into the generator	CRAFTS/GANG.js	HIGH (before any regenerate!)
7	Apply the price table from Step 2	MINING.js, DrugLab.js, WeaponDealer.js	MEDIUM
8	Remove dead weapon-part entries from the dealer	WeaponDealer.js	MEDIUM
9	Add automatic exploit check to the generator build	generate_economy_config.js	MEDIUM
10	Verify burger benches & junk-bin respawn are time-gated	backend	LOW

All numbers in this report were computed from your real config files (mining shop, supermarket, EMS shop, drug dealer, weapon dealer, craft recipes) and verified against the live server code paths named above.